Python Basic Synytax, and Variables

Variables and Assignment

In [10]:

```python
student_name = "John Doe"
age = 22
is_enrolled = True
gpa = 3.85

### Input / Ouput Function
print(f"Student: {student_name}")
print(f"Age: {age} | GPA: {gpa} | Enrolled: {is_enrolled}")
```

```
Student: John Doe
Age: 22 | GPA: 3.85 | Enrolled: True
```

Basic Data Types

- Interger : int
- Flort : float
- string : str
- Boolean : bool
- None Type : NoneType

In [12]:

```python
## Numeric Types
int_val = 42
float_val = 3.14159
print(f"Types: {type(int_val)}, {type(float_val)}")
```

```
Types: <class 'int'>, <class 'float'>
```

In [13]:

```python
### String Type
str_val = "Toyota"
str_val
```

Out[13]:

```
'Toyota'
```

In [14]:

```python
print(f"Types: {type(str_val)}")
```

```
Types: <class 'str'>
```

In [15]:

```python
### None Type
Missing_value = None
boolean_value = False

print(f"Types: {type(Missing_value)}, {type(boolean_value)}")
```

```
Types: <class 'NoneType'>, <class 'bool'>
```

Conversion of Data Types

str -> inteeger -> float -> str

In [19]:

```python
x = "100"
y = int(x)
z = float(y)
w = str(z)

print(f"Types: {type(x)}, {type(y)}, {type(z)}, {type(w)}")
```

```
Types: <class 'str'>, <class 'int'>, <class 'float'>, <class 'str'>
```

Mathematical and Comparison Operators

In [20]:

```python
a , b = 15, 4
print("Arithmetic:")
print(f"Add: {a+b} | Sub: {a-b} | Mul: {a * b}")
```

```
Arithmetic:
Add: 19 | Sub: 11 | Mul: 60
```

In [21]:

```python
print("Advanced Operation:")
print(f"Div: {a/b} | Floor: {a//b} | Mod: {a%b} | Pow: {a**4}")
```

```
Advanced Operation:
Div: 3.75 | Floor: 3 | Mod: 3 | Pow: 50625
```

Logical Operators and Work Flows

In [26]:

```python
score = 39

## Conditional Logic
if score >= 90:
    grade = "A"
elif score >= 75:
    grade = "B"
elif score >= 55:
    grade = "C"
elif score >= 40:
    grade = "D"
else:
    grade = "Fail"

print(f"Score: {score} -> Grade: {grade}")
```

```
Score: 39 -> Grade: Fail
```

Introduction to Pandas

In [ ]:

```
pip install pandas
pip install numpy
```

In [3]:

```python
import pandas as pd

### Create a dictionary
data = {
    "Name":["Alice", "Bob", "Charlie", "Diana"],
    "Age": [24, 30, 22, 28],
    "Score": [88.5, 92.0, 79.3, 95.1],
    "Department": ["Comp Sci", "Mathematics", "Physics", "Biological"]
}
data
```

Out[3]:

```
{'Name': ['Alice', 'Bob', 'Charlie', 'Diana'],
 'Age': [24, 30, 22, 28],
 'Score': [88.5, 92.0, 79.3, 95.1],
 'Department': ['Comp Sci', 'Mathematics', 'Physics', 'Biological']}
```

Convert the Dictionary to a Data Frame

In [4]:

```python
df = pd.DataFrame(data)
print("Created Data Frame")
df.head()
```

```
Created Data Frame
```

Out[4]:

	Name	Age	Score	Department
0	Alice	24	88.5	Comp Sci
1	Bob	30	92.0	Mathematics

	Name	Age	Score	Department
2	Charlie	22	79.3	Physics
3	Diana	28	95.1	Biological

Export this Data Frame to the Local Directory

In [5]:

```
df.to_csv("student_score.csv", index = False)
print("Created Sample CSV File Saved")
```

```
Created Sample CSV File Saved
```

Read / Import the File Back to Jupyter

In [8]:

```
df2 = pd.read_csv("student_score.csv")
df2.head()
```

Out[8]:

	Name	Age	Score	Department
0	Alice	24	88.5	Comp Sci
1	Bob	30	92.0	Mathematics
2	Charlie	22	79.3	Physics
3	Diana	28	95.1	Biological

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